

Problem L

Park of Three

Time limit: 1 second

Problem Description

The Park of Three is a famous landmark in the Kingdom of Three. It contains n magical pillars connected by m *magical connections*. The pillars are numbered from 1 to n . The i -th connection links pillar u_i and pillar v_i , and all connections are bidirectional.

Believing that the number three holds protective power against evil forces, the park's architects ensured that every pillar is incident to exactly three connections. Note that a connection may link a pillar to itself. In such cases, the self-loop counts twice toward the number of pillar's incident connections.

Triangles formed by the pillars are the source of their protective power. Historically, these triangles successfully repelled evil forces hundreds of years ago. To assess the network's true resilience, you must determine its structural vulnerability.

Formally, three distinct pillars i, j, k form a *triangle* if the connections (i, j) , (j, k) , and (k, i) all exist. When a pillar is destroyed, all connections incident to it are removed. The *strength* of the park is defined as the minimum number s such that destroying s specific pillars eliminates all triangles. Please determine the strength s and output the indices of s pillars such that there will be no triangles after destroying them.

Input Format

The first line contains an integer n , representing the number of magical pillars in the park.

Followed by $m = \frac{3n}{2}$ lines, the i -th of which contains two integers u_i and v_i , representing the endpoints of the i -th magical connection.

Output Format

On the first line, output the integer s .

On the second line, output s distinct integers representing the indices of the pillars to be destroyed. After removing these pillars and all incident connections, no triangles should remain.

If multiple solutions exist, output any of them. The indices may be printed in any order.

Technical Specification

- $4 \leq n \leq 2 \times 10^5$, and n is even.
- $1 \leq u_i, v_i \leq n$ for all $1 \leq i \leq m$.
- The network of magical connections may contain self-loops and multiple connections between the same pair of pillars.
- Each pillar is incident to exactly 3 connections, where each self-loop is counted twice toward this total.

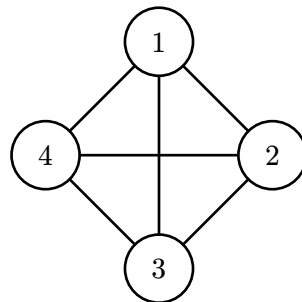
Sample Input 1

```
4
1 2
1 3
1 4
2 3
2 4
3 4
```

Sample Output 1

```
2
1 2
```

Explanation for Sample 1: The network of connections is shown in the figure below. There are 4 triangles formed by the pillars: $(1, 2, 3)$, $(1, 2, 4)$, $(1, 3, 4)$, and $(2, 3, 4)$. After destroying pillars 1 and 2, no triangle remains in the network. Any subset of $\{1, 2, 3, 4\}$ of size 2 is also a valid solution.



Sample Input 2

```
6
1 2
3 1
3 2
4 1
4 2
3 5
4 6
5 5
6 6
```

Sample Output 2

```
1
1
```